

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution uses Machine Learning to recommend the most suitable crop based on soil nutrients and environmental conditions. It helps farmers make informed decisions, improve crop productivity, and reduce the risk of crop failure.

Project Overview	
Objective	Develop a Machine Learning-based system that recommends the most suitable crop using soil nutrients (N, P, K), temperature, humidity, pH, and rainfall.
Scope	Covers data preprocessing, model training, prediction, evaluation, and deployment using Flask.
Problem Statement	
Description	Farmers often face difficulties in selecting the right crop due to varying soil and environmental conditions, resulting in lower productivity and financial losses.
Impact	The solution improves crop selection, enhances agricultural productivity, optimizes resource utilization, and supports sustainable farming practices.
Proposed Solution	
Approach	Collect agricultural data, preprocess the dataset, train the Machine Learning model, predict the most suitable crop, and display the recommendation through the Flask interface.
Key Features	•Crop Recommendation using ML. • Soil & Environmental Data Analysis •Flask-Based Web Application • Fast and Accurate Predictions • User-Friendly Interface